



Though a conventional power station could provide the needed power levels, it would be neither small nor lightweight, and would definitely not meet the mobility requirement. Some of the potential candidates for building the prime power source include advanced technology batteries, turbo-generators and advanced magneto-hydrodynamic (MHD) generators using superconducting circuitry.

A typical energy-storage method involves charging a bank of capacitors, or spinning a huge mechanical flywheel, or simply storing the energy in the form of a high-energy

for another series of bursts of pulsed power.

Target tracking and beam pointing

Unlike some other areas of particle beam weapon development such as the generation of high-energy particle beam, which need to address certain basic issues of science and technology, target acquisition and tracking and beam-pointing technologies required for a particle beam weapon are not unique to this class of directed-energy weapons. In this respect, particle beam weapon programme has immensely benefit-

weapons and neutral-particle beam through space in the case of exoatmospheric weapons. The particle beam must have an extremely precise path of propagation as it traverses the required distance to the target.

The propagation problem is more severe in the case of charged-particle beam through atmosphere as compared to that of neutral-particle beam through space. Propagation of a neutral-particle beam does not suffer from beam instability problems possibly encountered by a charged-particle beam while propagating through atmosphere.

One of the contributing factors is beam spreading, which causes increase in beam diameter and consequent decrease in energy density as it travels towards the target. Even with a small amount of beam divergence, the beam diameter may become appreciable enough to be unacceptable for longer ranges.

Two factors contribute to beam spreading: beam divergence imparted by the accelerator itself and present at the exit of the accelerator, and mutual repulsion of beam particles. While in the case of charged-particle beams both the factors are responsible for beam spreading, it would strictly originate from accelerator in the case of neutral-particle beams.

Even in the case of a neutral-particle beam propagating through atmosphere, the surrounding air molecules strip the neutral particles of the electrons and transform it into a charged-particle beam. This would further lead to undesirable beam divergence.

In the atmosphere, however, even if the beam particles were neutral, air molecules would strip the surrounding electrons quickly from the beam's neutral atoms, turning the beam into a charged-particle beam. The charged particles within the beam would then



Fig. 8: Large Hadron Collider

explosive that is released in a contained explosion. There are a number of other energy-storage and release mechanisms, each having its own set of advantages and disadvantages. The preferred mechanism depends upon the particle accelerator and also whether the beam weapon is endoatmospheric or exoatmospheric.

The pulse-forming network is used to shape the power pulse. In the case of an endoatmospheric particle beam weapon, one shot would have a burst of very-short-duration pulses with burst frequency in the range of kHz. The prime power source may deliver power for a series of bursts of pulsed power followed by the beam weapon going to quiescent state while the energy storage device gets recharged

ted from high-energy laser weapon programme.

Notwithstanding commonality of technologies between the particle beam weapons and high-energy laser weapons, there do remain some tracking and pointing problems that need to be solved. Many of these arise out of propagation of charged-particle beam through the atmosphere and neutral-particle beam through space.

Beam propagation

Other than the high-energy particle beam source and target-tracking and beam-pointing technologies, third important element determining the success of a particle beam weapon is the propagation of the charged-particle beam through atmosphere in the case of endoatmospheric